

Receptive Field, Not Attention: What Improves Lightweight Violence Detection

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Abstract. Violence detection runs on every camera of a surveillance installation, so the deployed model has to be small, and the literature reports steady gains from cheap modules added to small video networks. We measure which of these gains hold up. On RWF-2000, Hockey Fights and RLVS we remove duplicated clips, keep clips of the same scene in one fold, take every decision on out-of-fold predictions and score each test split once. Three results follow. First, the person-detection crop that this line of work applies before the network costs 2.97 out-of-fold points ($p = 0.0002$); decoding and resizing the video, with no detector, is both the most accurate and the cheapest pipeline. Second, enlarging the channel-wise kernels of every stage of X3D-M to 5×5 , trained as a zero-initialised ring that fuses into one kernel for inference, adds 2.63 test points over three seeds, while the same widening in the first two stages only adds 0.75. Third, a motion-attention gate adds nothing, and the same gate fed an all-zero motion map performs as well. Averaging the weights of the four fold models that the protocol trains anyway gives one model as accurate as their ensemble. The final detector reaches 91.6% on RWF-2000 with 3.3M parameters and 5.9 GFLOPs per clip.

Keywords: Violence detection · Efficient 3D CNN · Receptive field
· Structural reparameterisation · Evaluation protocol.

1 Introduction

Surveillance cameras are installed at a scale that human operators cannot watch: an estimated 770 million were in use by 2019 [5], and an observer’s ability to notice rare events declines within twenty minutes of continuous monitoring [22]. Automatic violence detection flags the few streams that need attention. Because it runs on every camera, the model must be cheap. The most accurate models on RWF-2000 [4] are far from that budget: CUE-Net [25] reports 94.0% with 354M parameters, and video transformers report 91.8–92.4% with 67M to 305M parameters [29,24]. A second line of work [4,17,14,32] stays within a few GFLOPs and reports 87–90%, usually by adding a module to a compact backbone: a motion branch, an attention block, or a wider kernel.

This paper asks how much such additions contribute once they are measured carefully. We study two of them on X3D-M [12]: a Motion Attention gate that

rescales features with a multi-scale frame difference, and a reparameterised wide kernel that enlarges the channel-wise convolutions to 5×5 and fuses away at inference. The benchmarks do not make such a measurement easy. RWF-2000 has no validation split, so checkpoints are often chosen on the test split; nine of its training clips are copies of test clips; and single training runs are noisy enough to produce significant-looking differences by chance. We therefore remove duplicates, keep same-scene clips in one fold, take every decision on four-fold out-of-fold (OOF) predictions, repeat the key comparisons over three seeds, and score each test split once.

Under that protocol the two modules behave very differently. Motion Attention adds nothing, and a control that feeds the gate an all-zero motion map does as well as the real one, so the gate does not use the signal it is built on. The wide kernel helps, but only once it is applied to all four stages rather than the first two: most of its gain comes from the late stages, where the feature maps are small and a 5×5 kernel covers much of the scene. The largest effect, however, lies outside the network. The pipeline that the task inherits detects people and crops the video to them before the network sees it. Rebuilding RWF-2000 from the raw videos seven times shows that this crop costs about three points, and that the best pipeline has no detector in it. Our contributions are:

- A controlled study of the input pipeline, showing that the person-box crop costs 2.97 OOF points and that the most accurate pipeline is also the cheapest (Sect. 5.1).
- A study of where and how far to widen the kernels of a compact 3D network, with a gain of 2.63 test points that holds over three seeds and fuses away at inference (Sect. 5.2).
- A negative result for motion gating with the control that explains it (Sect. 5.3).
- A uniform soup of the fold models, which turns the models trained for an honest estimate into one deployable model as accurate as their ensemble, replicated on nine experiments (Sect. 5.4).

2 Related Work

Violence detection. Early methods encoded motion with dense trajectories [31], flow magnitude [15] or acceleration [1]. Two-stream networks [26] add optical flow, often at a higher cost than the network itself; 3D convolutional networks [30,3,13,12] learn from RGB alone, and video transformers [2,20] rely on large-scale pre-training. On RWF-2000, the Flow Gated Network [4] reaches 87.25%, SepConvLSTM [17] 89.75%, a two-stream multi-dimensional network [14] 89.70% below one million parameters, and an EfficientNet with bidirectional motion attention and temporal shift [32] 90.25%. CUE-Net [25] and our earlier configuration first detect people and crop the frame to them. Most of these works report accuracy on the official test split without stating how the checkpoint was chosen.

Efficient backbones and large kernels. X3D [12] expands a small network one axis at a time; X3D-M reaches 76.0% on Kinetics-400 with 3.8M parameters.

Structural reparameterisation [8,6] trains extra branches and fuses them into one kernel for inference, and RepLKNet [7,9] uses it to train very large kernels in 2D networks. We use the same principle on the channel-wise 3D kernels of a pre-trained video network, where the question is less how large a kernel can be than where a modest enlargement pays off.

Gating and weight averaging. Multiplicative gates such as squeeze-and-excitation [16] re-weight features with a learned signal. Model soups [33] average the weights of models fine-tuned from the same initialisation and often match their ensemble; we apply the idea to the fold models of a cross-validation protocol.

3 Method

Backbone. A clip is an RGB tensor of 16 frames at 224×224 . X3D-M [12], pre-trained on Kinetics-400 [19], has a stem and four stages (Res2–Res5 with 3, 5, 11 and 7 blocks) whose outputs are 56^2 , 28^2 , 14^2 and 7^2 at full temporal resolution. Each block holds a $3 \times 3 \times 3$ channel-wise convolution between two $1 \times 1 \times 1$ projections. The head pools globally and ends in a new two-way classifier.

Wide-kernel reparameterisation. Let W_{base} be the pre-trained $3 \times 3 \times 3$ channel-wise kernel of a block. It is replaced by

$$W_{\text{eff}} = \mathcal{P}(W_{\text{base}}) + M_{\text{ring}} \odot \Delta, \quad (1)$$

where $\mathcal{P}$ zero-pads the spatial dimensions to 5×5 , Δ is initialised to zero and M_{ring} is one on the 16 outer positions of each 5×5 slice. At initialisation the network is exactly the pre-trained one. Convolution is linear in its kernel, so after training the two terms are summed into one $3 \times 5 \times 5$ kernel and inference runs a plain X3D with slightly larger kernels; the fused and unfused networks agree to 4×10^{-7} in output probability. Keeping Δ as its own tensor has two consequences. It is trained at the learning rate of new parameters rather than that of the backbone, and it can be trained in a stage whose pre-trained weights stay frozen: the recipe below keeps Res4 frozen, and the ring still widens it. We apply the ring to every block of every stage (Fig. 1), which adds 0.29M parameters (0.46M during training) and 1.19 GFLOPs.

Motion Attention. The module we compare against computes a motion map $m_t = \min(\max(d_t^{(1)}, d_t^{(2)}), \delta)/\delta$ from the channel-averaged absolute frame differences at steps 1 and 2 ($\delta = 0.2$), turns it into four temporal modes with a learned temporal convolution, and maps them to a gain on the Res4 features,

$$Y = F \odot (1 + G), \quad G = \tanh\left(W_2 \text{ReLU}(W_1 \tilde{M}) + b_2\right), \quad (2)$$

with W_2 initialised near zero, so the module starts as an identity. It varies the gain over channels, time and position and has 2,508 parameters.

Fold soup. The protocol of Sect. 4 trains four models, one per fold, from the same pre-trained weights. Instead of retraining on the whole training split

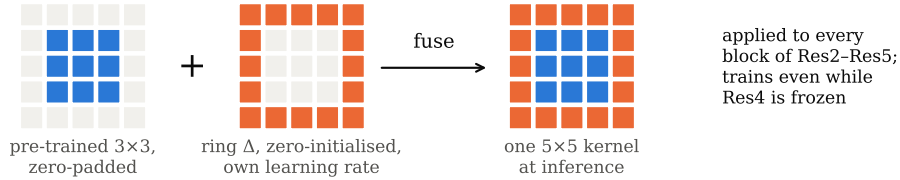


Fig. 1. The reparameterised wide kernel. The pre-trained 3×3 kernel is zero-padded to 5×5 and a zero-initialised ring Δ is added (Eq. 1); after training the two are summed into one kernel, so the deployed network is a plain X3D with 5×5 channel-wise kernels.

or deploying their four-model ensemble, we average their weights uniformly into one network. The recipe is fixed in advance, every fold with equal weight, so scoring it on the test split involves no selection.

Training. In a probe phase of four epochs at 10^{-3} only the classifier and the new modules train. In the fine-tuning phase Res2, Res3, Res5 and the head train at 5×10^{-5} and new parameters, including the rings, at 5×10^{-4} ; the stem and Res4 stay frozen. We use AdamW [21] with weight decay 5×10^{-4} , a cosine schedule, label smoothing [28] of 0.1, CutMix [34] on half of the batches, batches of 12 clips, 48 epochs with patience 16 and an exponential moving average of the weights. Augmentation draws one set of parameters per clip (photometric changes, flip, crop of 70–100%, rotation of $\pm 10^\circ$, playback speed) and applies it to every frame. At test time 16 frames are spread evenly over the clip, one view.

4 Evaluation Protocol

Datasets. RWF-2000 [4] has 2,000 five-second clips of fights and non-violent activity, mostly from surveillance cameras, split into 1,600 training and 400 test clips. Hockey Fights [1] has 1,000 clips from ice-hockey games and RLVS [27] 2,000 clips of street fights and everyday activities; neither has an official split, so we draw 20% of each as a test set by whole same-scene groups. Every clip is stored as 64 frames.

Duplicate audit. Each clip is described by four evenly spaced frames, converted to grey, pooled to 16×16 and standardised; two clips are compared by the cosine similarity of their descriptors. Pairs above 0.99 are treated as copies and pairs above 0.9 as the same scene. In RWF-2000, nine training clips are copies of test clips; with the second copies of duplicated training pairs, we remove 16 training clips, leaving 1,584 in 1,579 same-scene groups. Hockey Fights and RLVS keep 797 and 1,588 training clips after the same audit.

Decisions and statistics. None of the benchmarks has a validation split. We divide each training split into four folds, stratified by class and never separating a same-scene group, and keep the checkpoint with the best validation F1 in each. The four validation sets cover every training clip once, which gives one

Table 1. RWF-2000 rebuilt from the raw videos, one preprocessing step at a time; same network, recipe and folds, seed 0. Δ and p are against the original pipeline (row 6), exact McNemar on the OOF predictions.

Crop	Contrast	Resize	OOF	Test	Δ OOF	p
none	CLAHE	stretch	92.55	89.50	+2.97	0.0002
none	–	stretch	92.42	89.75	+2.84	0.0003
none	CLAHE	pad	92.36	88.75	+2.78	0.0008
$\geq$ half frame	CLAHE	stretch	91.98	89.50	+2.40	0.0007
person cluster	–	stretch	90.40	87.00	+0.82	0.29
person cluster	CLAHE	stretch	89.58	88.50	–	–
union of people	CLAHE	stretch	89.39	90.50	–0.19	0.85

OOF prediction per clip; every choice in this paper is made on these predictions, and the test split is scored once. Unless stated otherwise, test accuracy is the mean over the four fold models, each trained on 75% of the training clips. Paired comparisons use the exact McNemar test [23] and 95% bootstrap intervals over clips [10]. On the 1,584 OOF clips of RWF-2000 the half-width of this interval is about 1.1 points; on its 400 test clips one point is four clips, which is why the central comparisons use three seeds.

5 Experiments

5.1 The input pipeline

The pipeline of our earlier configuration detects people with YOLOv8n [18] on 12 frames per clip, clusters the box centres with DBSCAN [11], crops every frame to the largest cluster with a 20% margin, equalises contrast with CLAHE [35] and resizes to 224×224 . We rebuilt RWF-2000 from the raw videos seven times, changing one step at a time, and trained the same network (X3D-M with Motion Attention and wide kernels in Res2–Res3) with the same recipe on the same folds (Table 1).

Removing the crop is worth +2.97 OOF points ($[+1.45, +4.48]$, $p = 0.0002$). A gentler crop that keeps at least half of the frame recovers most of the loss but does not beat the uncropped frame, and the union of all person boxes, as used by CUE-Net [25], is the worst of the seven. Once the frame is left intact, contrast equalisation (-0.13 , $p = 0.90$) and keeping the 16:9 aspect ratio by padding instead of stretching (-0.19 , $p = 0.84$) make no measurable difference, so the best pipeline is also the cheapest: decode and resize, with no detector. The errors of the cropped model point to the reason. The clips it misclassifies hold people who occupy 4.0% of the frame against 7.5% in the clips it gets right, and no person is detected at all in 49% of them; cropping towards small or undetected actors removes the context that separates a fight from a crowd. The union row also shows why the decision is made out of fold: it has the best test accuracy of the seven pipelines and the second worst OOF accuracy, and choosing on the test

Table 2. Where and how to widen, RWF-2000, uncropped pipeline, seed 0. Test: mean of the four fold models. Parameters as trained; GFLOPs per view.

Kernel	Stages	Params	GFLOPs	OOF	Test
none (X3D-M)	–	2.98M	4.73	92.80	88.37
5×5	Res2–Res3	3.03M	5.45	92.49	89.44
$5 \times 5 / 7 \times 7$	Res2–Res5	3.83M	6.63	92.99	89.44
$5 \times 5 \times 5$	Res2–Res5	3.74M	7.15	92.55	90.06
7×7 , dilated	Res2–Res5	3.14M	7.70	92.93	90.31
5×5	Res4–Res5	3.38M	5.20	92.61	90.75
5×5	Res2–Res5	3.44M	5.92	92.99	91.25

Table 3. X3D-M with and without 5×5 kernels in every stage, three seeds of four folds. Test: mean of the twelve fold models. Parameters after fusing the kernels.

Model	Params	GFLOPs	OOF (mean of 3)	Test
X3D-M	2.98M	4.73	92.59	88.18
5×5 in Res2–Res5	3.27M	5.92	93.20	90.81
Difference			+0.61	+2.63

split would have cost three points. All following experiments use the uncropped pipeline.

5.2 Where and how far to widen the kernels

Table 2 varies where the kernels are widened and their shape. The original configuration widened Res2–Res3, on the grounds that early stages hold the large feature maps; there it adds little over X3D-M. Widening all four stages gives the best result, and widening only Res4–Res5 gives most of it. At Res4 and Res5 the feature maps are 14×14 and 7×7 , so a 5×5 kernel covers a large part of the scene. Larger or differently shaped kernels do not help: a 7×7 receptive field through dilated reparameterisation [9], kernels that are also wider in time, and graded $5 \times 5/7 \times 7$ kernels all fall between X3D-M and the plain 5×5 kernel. Out of fold all variants are within noise (92.49–92.99); the ranking is by test accuracy.

Because single seeds have misled us before (Sect. 5.6), we repeated X3D-M and the all-stage 5×5 kernel over three seeds (Table 3). The gain is the same size in every seed, +2.87, +2.38 and +2.62 test points, and exact McNemar on the fold ensembles gives $p = 0.0043$, 0.17 and 0.0002. It is larger on the test split than out of fold in every seed (+2.63 against +0.61). The OOF clips share their scenes with the training folds, whereas RWF-2000 draws its test clips from different videos, so a change that mainly helps on unfamiliar scenes looks smaller out of fold; the pipeline result behaves in the same way. We report both and keep the out-of-fold number as the conservative one.

Table 4. Motion Attention on the uncropped pipeline, seed 0. Δ : test accuracy of the fold ensemble against X3D-M, exact McNemar p .

Model	Params	OOF	Test (ens.)	Δ	p
X3D-M	2.98M	92.80	89.50	–	–
+ Motion Attention	2.98M	92.61	88.00	−1.50	0.11
+ MA + 5×5 in Res2–Res3	3.03M	92.55	89.50	0.00	1.00
+ 5×5 in Res2–Res5 (ours)	3.44M	92.99	93.00	+3.50	0.0043

5.3 Motion attention

Table 4 measures Motion Attention on the uncropped pipeline. On its own it is the weakest variant we trained, and adding it to the wide kernels does not change them. The reason is visible in a control run on the original pipeline: the same gate, with every parameter kept but fed an all-zero motion map, reached 91.09 OOF against 90.65 for the real gate and 90.84 for X3D-M. Whatever the gate contributes, it does not come from the motion signal it is built around. The earlier gain of the two modules together (+0.88 OOF over three seeds) appeared only under a conventional fine-tuning recipe with a backbone learning rate of 10^{-5} and disappeared once the recipe was tuned (-0.04 , $p = 0.78$).

5.4 One model from the folds

The four fold models of every experiment form an ensemble that is more accurate than any one of them, but at four times the inference cost. Table 5 compares a single fold model, the ensemble and the uniform soup on nine experiments over three datasets, eight of which were not used while developing the soup. Against a single model the soup gains 0.69 points on average, with seven wins, no losses and two ties (sign test $p = 0.016$); against the ensemble it is level on average, at a quarter of the cost. The gain follows the spread between fold models: it is largest on RWF-2000 and vanishes on RLVS, where accuracy is already 98.5%. Re-estimating the BatchNorm statistics after averaging helped on the first experiment but won four and lost four of the other eight, so it is not part of the recipe.

5.5 The final model and comparison

The final detector is X3D-M with 5×5 kernels in all four stages, trained on the uncropped pipeline and deployed as the soup of its four fold models. Over three seeds it reaches 91.58% test accuracy (91.50, 92.00, 91.25) with 3.27M parameters and 5.92 GFLOPs per clip (Fig. 2). Our earlier configuration, with the cropped pipeline, Motion Attention and wide kernels in Res2–Res3, reaches 86.50% under the same protocol when trained on the whole training split; it had been reported at 90.5% with the epoch selected on the test split.

Table 5. Test accuracy of a single fold model (mean), the four-model ensemble and the uniform fold soup. (3s): mean over three seeds.

Model	Data	Single	Ensemble	Soup
5×5 in Res2–Res5 (3s)	RWF-2000	90.81	92.42	91.58
X3D-M (3s)	RWF-2000	88.19	88.92	89.00
MA-X3D, uncropped	RWF-2000	89.06	89.50	90.50
MA-X3D, no CLAHE	RWF-2000	89.38	89.75	90.25
MA-X3D, adaptive crop	RWF-2000	87.88	89.50	89.25
X3D-M (3s)	Hockey	95.62	96.00	95.67
MA-X3D (3s)	Hockey	95.46	96.17	96.00
X3D-M	RLVS	98.49	98.49	98.49
MA-X3D	RLVS	98.17	98.49	98.49

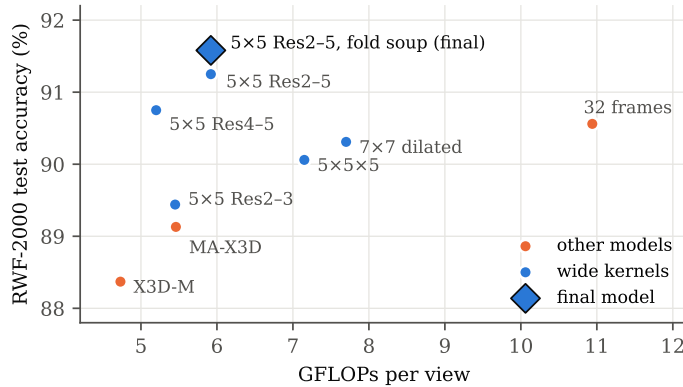
**Fig. 2.** Test accuracy against computation on RWF-2000, uncropped pipeline. Circles: mean of the four fold models, seed 0. Diamond: the final model, fold soup, mean of three seeds.

Table 6 places the result among published models. The published numbers are not strictly comparable, since they are obtained on the official split, which contains the duplicated clips, usually without stating how the checkpoint was chosen; the selection alone is worth about 0.8 points (Sect. 5.6). Under our protocol the model is the most accurate below 5M parameters and trails a VideoMAE-B transformer, trained on the same pipeline, by 2.4 points at a twenty-fifth of its parameters.

5.6 What the protocol is worth

Two effects outside the model are of the same size as the modules we tested. Logging test accuracy after every epoch, choosing the epoch by test accuracy instead of validation F1 inflates the reported number by 0.8 points on average on

Table 6. RWF-2000 test accuracy. Top: as reported, official split, selection procedure mostly unstated. Bottom: this paper’s protocol. [†]Trained on RWF-2000, Hockey Fights and RLVS.

Method	Params	GFLOPs	Test acc.
Flow Gated Network [4]	–	–	87.25
SepConvLSTM [17]	–	–	89.75
MDCN [14]	<1M	–	89.70
BiLTMA [32]	4.48M	–	90.25
VideoMAE [29,24]	67–305M	–	91.8–92.4
CUE-Net [25]	354M	~5800	94.00
X3D-M, uncropped (3 seeds)	2.98M	4.73	88.18
VideoMAE-B, uncropped [†]	86.2M	135	94.00
Ours: 5×5 Res2–Res5, fold soup (3 seeds)	3.27M	5.92	91.58

RWF-2000. And on Hockey Fights, one seed gave the two modules a significant +1.25 OOF points ($p = 0.006$) that three seeds turned into -0.04 . Under the usual practice, either effect alone could produce a published improvement.

6 Conclusion

We measured what improves a compact violence detector under a protocol that removes duplicated clips, keeps same-scene clips together, decides out of fold and scores each test split once. The largest effect is in the input pipeline: cropping the video to detected people costs about three points, and the best pipeline uses no detector. Inside the network, widening the channel-wise kernels of every stage to 5×5 adds 2.63 test points over three seeds and fuses away at inference, with most of the gain in the late stages, while a motion-attention gate adds nothing and does not use its motion input. Averaging the fold models that the protocol trains anyway gives a single model as accurate as their ensemble. The resulting detector reaches 91.6% on RWF-2000 with 3.3M parameters and 5.9 GFLOPs. Two limitations remain: most comparisons beyond the central one use a single seed, and the pipeline study covers RWF-2000 only. We release the splits, the duplicate audit, the rebuilt pipelines and the training code.

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